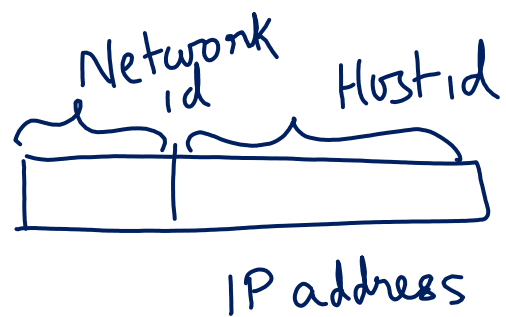
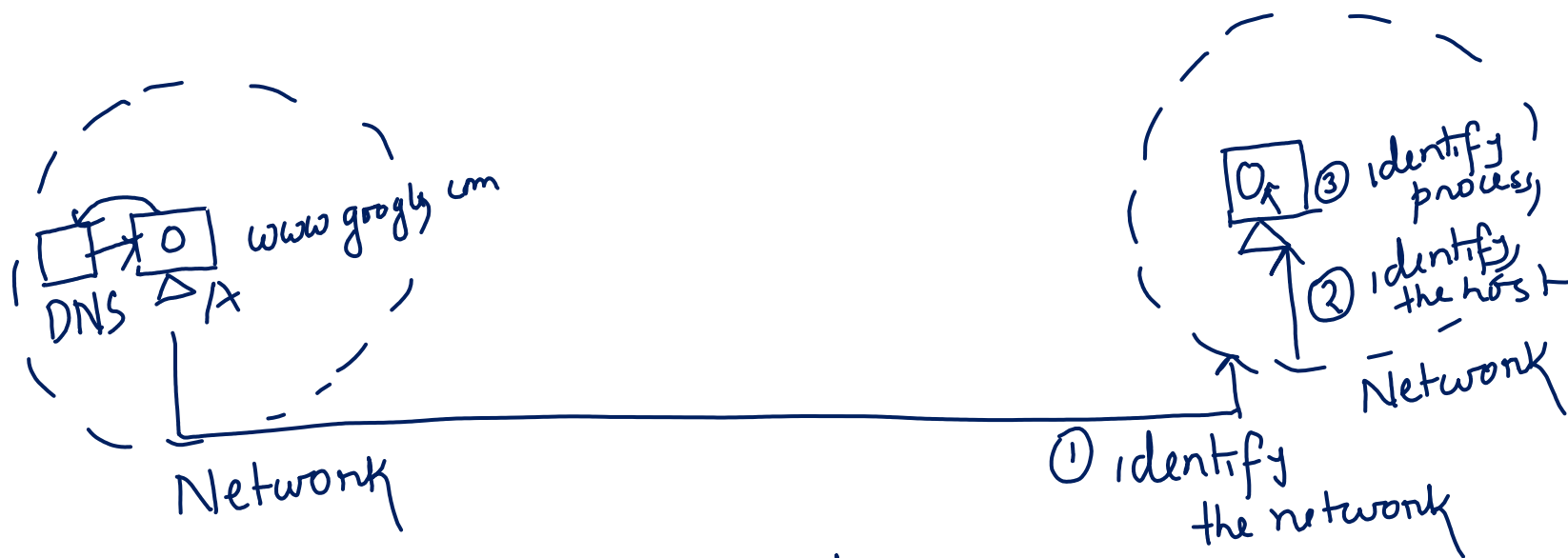


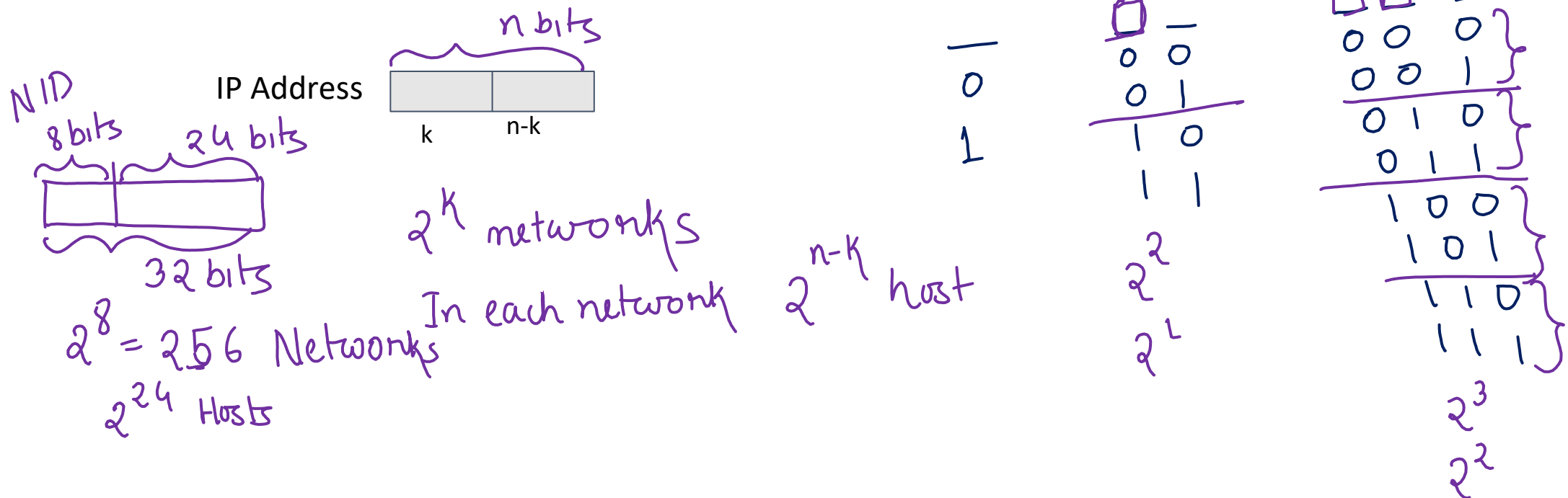
IP Addressing

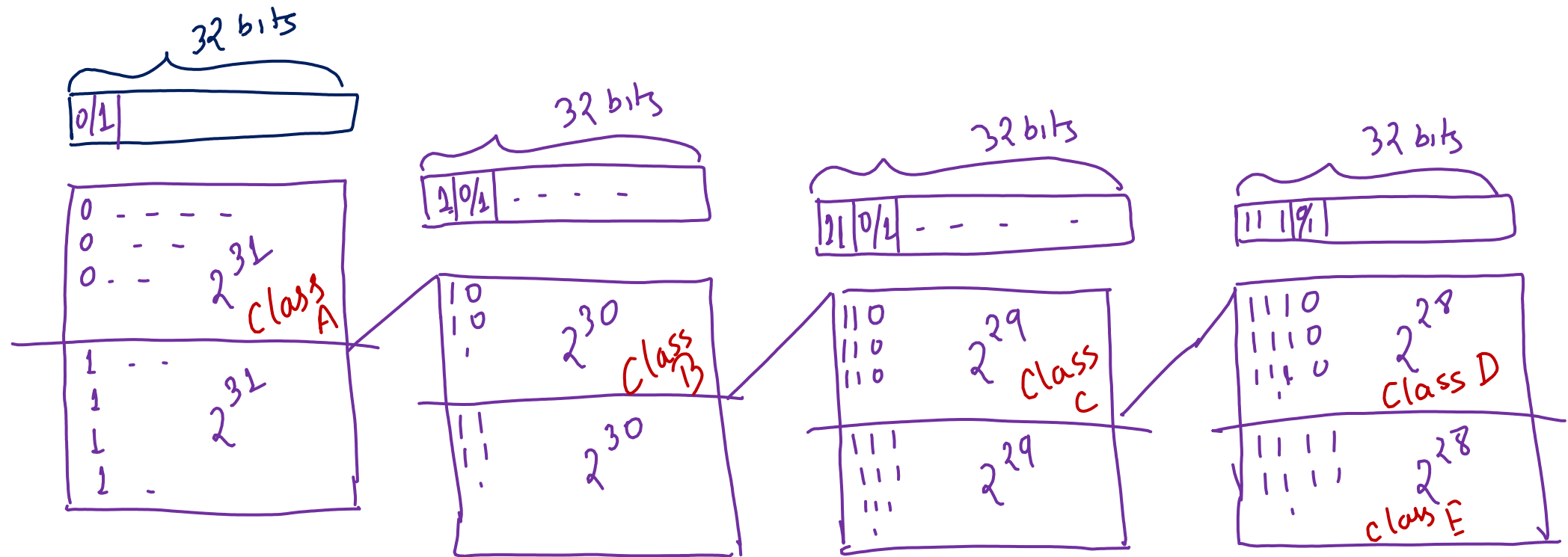
-Dr. Sumana Maiti



- With 1 bit 2 binary numbers are possible {0,1}
- With 2 bits, 4 binary numbers are possible {00, 01, 10, 11}
- With 3 bits, 8 binary numbers are possible {000, 001, 010, 011, 100, 101, 110, 111}
- From n bits, 2^n numbers are possible. If I choose k bits from n bits, then we can get 2^k parts.

One part $2^n/2^k = 2^{(n-k)}$ numbers.

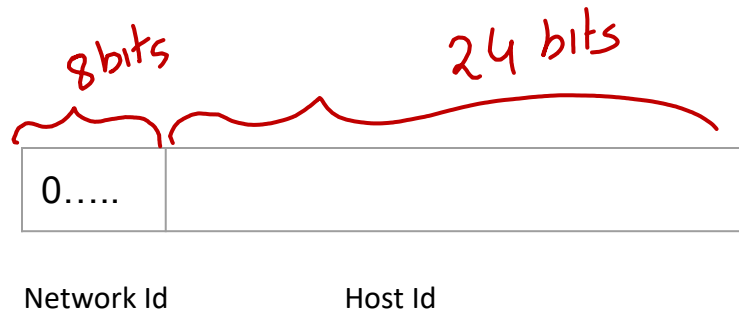




$$\underbrace{(0011 \dots)}_{8 \text{ bit}} \cdot \underbrace{(\dots)}_{8 \text{ bit}} \cdot \underbrace{(\dots)}_{8 \text{ bit}} \cdot \underbrace{(\dots)}_{8 \text{ bit}}$$

Class A

Number of IP addresses in class A - 2^{31}



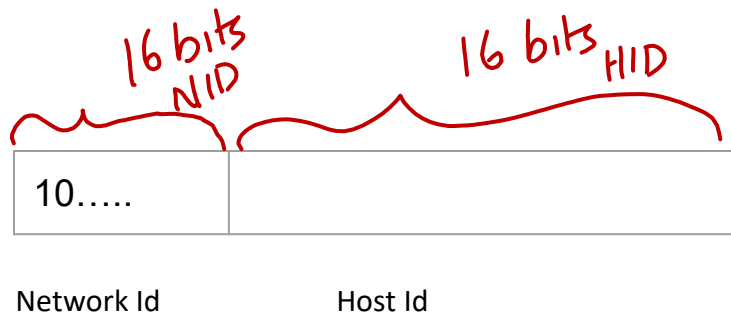
- 8 bit Network id
- 24 bit host id
- $2^7 = 128$ Network of CA
- $2^{24} = 16$ million IP
- 0 to 127 are networks, (0 and 127 are excluded)

0 - - - - -
0 0 0 0 0 0 0 - 0
0 0 0 0 0 0 1
0

1 1 1 1 1 1 1 - 127

Class B

Number of IP addresses in class B - 2^{30}

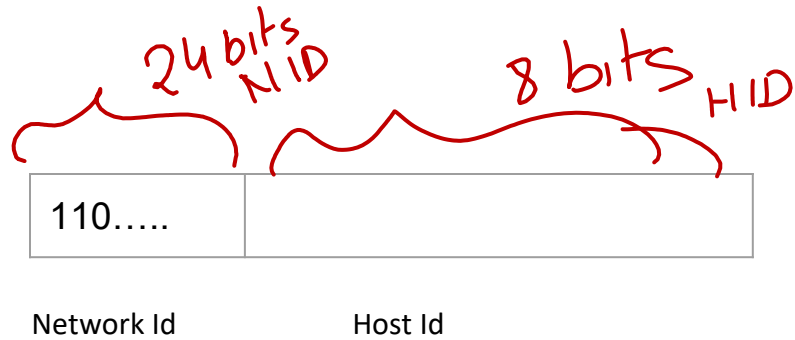


- 16 bit Network id
- 16 bit host id
- 2^{14} Networks of ~~CA~~ CB
- 2^{16} IP addresses for each network
- Range of class B 128 to 191

10 — — — — — — — 128
0 0 0 0 0 0
0 0 0 0 0 1
:
1 1 1 1 1 1 — 191

Class C

Number of IP addresses in class C - 2^{29}



- 24 bit Network id
- 8 bit host id
- 2^{21} Networks of C ~~A~~
- 2^8 IP addresses for each network
- Range of class B 192 to 223

110
— — — — —
0 0 0 0 0 — 192
0 0 0 0 1
:
1 1 1 1 1 — 223

Class D and Class E

Range of Class D : 224 to 239

Range of Class E: 240 to 255

Types of Casting

Two types:

- **Unicast:** Send a packet from one host to one host
- **Broadcast:** Send a packet from one host to multiple hosts.

Broadcast

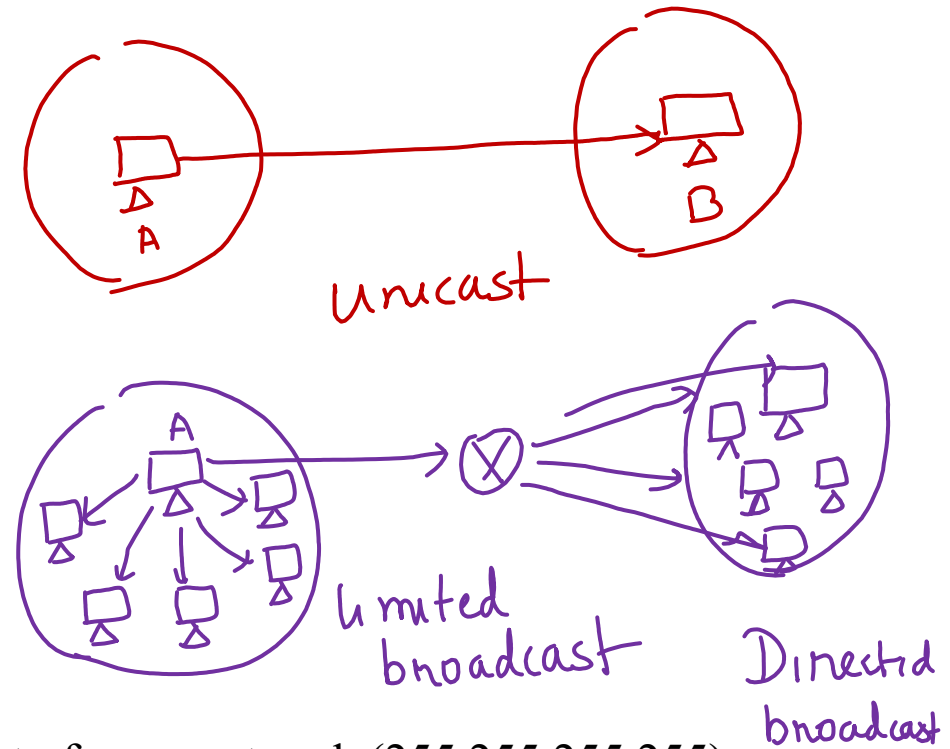
- ❖ **Limited Broadcast:** Send a packet to all the host of same network (255.255.255.255)
- ❖ **Directed Broadcast:** Send a packet to all the host of other network (20.255.255.255)

last ip address
of each network
is directed
broadcast
id

first ip address
of each network is
network id

m	Source ip	Dest ip
m	10.1.2.3	255.255.255.255

15 1 0 0
15 0 0 0
15.255.255.255



Q1. Find the network id, directed broadcast id, and limited broadcast id from the following IP?

1) 10.15.20.60

2) 130.1.2.3

3) 200.1.10.100

1) $\frac{10}{\downarrow \text{NID}}$ $\frac{15 \ 20 \ 60}{\downarrow \text{HID}}$ - class A
 { 8 bits - NID
 24 bits - HID

Network id - 10 0 0 0

Directed broadcast id - 10.255 255 255

Limited broadcast id - 255 255 255 255

2) $\frac{130 \ 1}{\downarrow \text{Network id}}$ $\frac{2 \ 3}{\downarrow \text{Host id}}$ - class B
 16 bits - NID
 16 bits - HID

Network id - 130 1. 0 0

Directed broadcast id - 130 1. 255 255

Limited broadcast id - 255 255 255 255

3) $\frac{200 \ 1 \ 10}{\text{NID}}$ $\frac{100}{\text{HID}}$ - class C
 24 bits - NID
 8 bits - HID

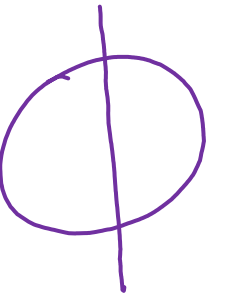
Network id - 200.1 10 0

Directed broadcast id - 200.1.10. 255

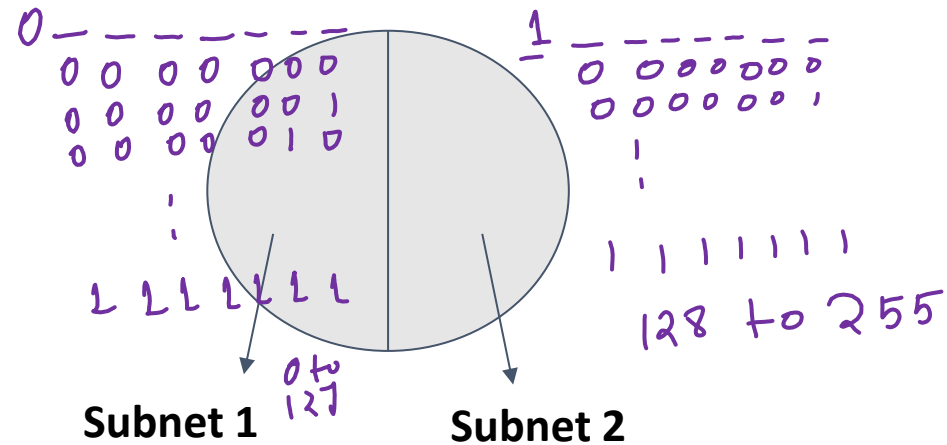
Limited broadcast id - 255 255 255 255

Subnet

- We divide the network into small network, which is called **Subnetting**
- Advantages:
 - Maintenance is easier
- Disadvantages
 - For **big network identification** of the process needs **three steps**
 - *Identify network*
 - *Identify host*
 - *Identify process*
 - For **subnetting**, the **identification is complex**
 - *Identify network*
 - *Identify subnet*
 - *Identify host*
 - *Identify process*



Example: For class C network 200.1.2.0

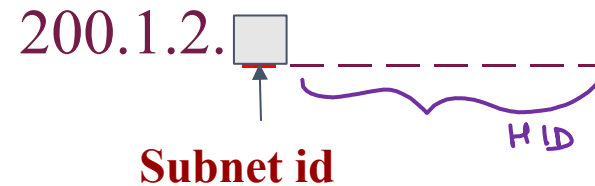


For Subnet 1
200.1.2.0: Network id
200.1.2.127: Directed
broadcast id

For Subnet 2
200.1.2.128: Network id
200.1.2.255: Directed
broadcast id

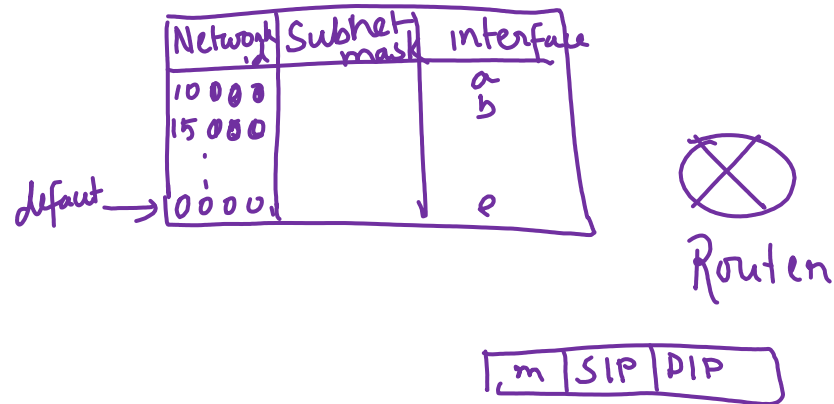
Host id: 0/1 - - - - -

*First bit of host id is chosen for
subnetting*



Subnet Mask

- Router needs to choose the network in which the particular IP address belongs
- To solve this, Subnet Mask is used.
- Subnet mask is 32 bit number, contains 0 and 1
- The number of 1 represent Network id, and number of 0 represent Host id



255 0 0 0 — class A
255 255 0 0 — class B
255 255 255 0 — class C

Example:

Subnet mask: $\overbrace{11111111 . 11111111 . 11111111 . 11}^{\text{NID}} \underbrace{0000000}_{\text{HID}}$

Let IP address is 200.1.2.130

AND the IP address with the Subnet mask. The result will be the network id

AND

11001000 . 00000001 . 00000010 . 10000010	
<u>11111111 . 11111111 . 11111111 . 11000000</u>	
11001000 . 00000001 . 00000010 . 10000000	= 200.1.2.128
	<u>NID</u>

- Let destination IP address is 200.1.2.0. In which interface the router will send the packet?

Network id	Subnet Mask	Interface
200.1.2.0	255.255.255.192	a
200.1.2.64	255.255.255.192	b
200.1.2.128	255.255.255.192	c
200.1.2.192	255.255.255.192	d
0.0.0.0	0.0.0.0	e

11001000.00000001.0000010
 00000000
 AND
 11111111 11111111.11111111
 11000000
 11001000.00000001 0000010
 00000000
 200.1.2.0

Classless inter domain routing

- In Classfull addressing, there is no flexibility
- Classless addressing solves this
- Instead of Network id, we write block id here
- a.b.c.d/n Here, n indicates the number of bits for network id part

Example: *20.10.50.100/20*

The number of bits used for NID part is 20

Network id: 20 bits

Host id: $32-20=12$ bits

Rules for CIDR block

- All IP address should be contiguous
- The block, asked by user must be 2^n . User cannot ask random blocks.
- First IP address in the block should be evenly divisible by size of the block.

Q1. Check whether a given IP address is CIDR block or not?

100.1.2.32 to 100.1.2.47?

- ✓ All IP addresses are contiguous
- ✓ 16 IP addresses It can be represented by 2^4
- ✓ 100 | 2 00 10000

- All the IP addresses are contiguous
- $\text{Size} = 1 + (47 - 32) = 16 = 24$
- First IP address is evenly divisible by the size of the block
100.1.2.00100000